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Evaluation of the physiochemical and metabolite of different region coffee beans by using UHPLC-QE-MS untargeted-metabonomics approaches

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ABSTRACT

Coffee is one of the most important agricultural commodities and has the unique organoleptic characteristics such as strong but not bitter taste, fragrant, oily, and fruity. In this study, an untargeted metabolomics approach based on UHPLC-QE-MS was used to investigate the differences in terms of components of precursor metabolites in coffee beans from 18 producing regions worldwide. Fingerprint analysis, principal component analysis and hierarchical clustering analysis revealed a neat separation among coffee beans. Compounds with high relevance to variance in the projection values in supervised multivariate analysis were selected as important metabolites for the discrimination of coffee samples. In total, 10 different families of compounds were considered as potential markers of the coffee beans: 3-hydroxycoumarin, 4,5-di-O-caffeoylquinic, cryptochlorogenic acid, palmitic amide, linoleamide, arachidic acid, petroselinic acid, trehalose, L-glutamic acid, L-malic acid. The findings presented herein serve as a suitable framework for the design of novel discrimination strategies in food origin tracing.

1. Introduction

Coffee is one of the world's most important tradable products of long supply chain and high economic value (Verhage et al., 2017). Currently, coffee is grown in 79 countries and regions around the world. According to reports released by the Food and Agriculture Organization of the United Nations (FAO) (FAO, 2020), 18 countries of each has harvested areas more than 100,000 hectares of coffee plantations, and 15 countries of each was responsible for the production of over 100,000 tons of coffee in 2018/19, among which are included Brazil, Ethiopia, Mexico, and China. And in China, green coffee was produced in Yunnan Province more than 98%. Furthermore, coffee consumption has been increasing annually, having raised by 3.25% over the previous year, reaching 9.8857 million tons in 2018/19, according to data published by the United States Department of Agriculture (USDA) (USDA, 2020). *Arabica* (*Coffea arabica* L.), *Robusta* (*Coffea canephora*) and *Liberica* (*Coffea liberica*) are the main species of coffee cultivated wordwide, among which *Arabica* is best known for possessing the most desirable qualities, flavor and aroma (Cong et al., 2020) Currently, *Arabica* and *Robusta* are the common coffee beans in the market.

Coffee contains a great variety of biologically active compounds, including alkaloids (caffeine, theobromine, theophylline, trigonelline), polyphenols (chlorogenic acid, flavonoids, coumarins, lignans), carbohydrates, amino acids, proteins and lipids (R Pérez-Míguez et al., 2020). Such biologically-active coffee compounds may prevent neurodegenerative diseases and depression (Dam et al., 2020), as well as modulate gut microbiome (Jaquet et al., 2009) and glucose (Vitaglione et al., 2010) and fat metabolism hence alleviating metabolic syndromes (Lecoultre et al., 2014), whilst maintaining both antioxidant and anti-inflammatory activities (Lara-Guzmán et al., 2020). Furthermore, coffee active compounds are also precursors of flavor in coffee. However, the quality of coffee beans is significantly impacted by conditions of the environment, crop management, harvest and post-harvest in the different production regions (Larissa et al., 2016). Therefore, it is crucial for studies focusing on coffee authentication to distinguish the variations in chemical composition of coffee from different regions of the

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world.

Metabolomics is an important part of systems biology which comprises quantitative and qualitative measurements of metabolites produced by a living organism or cell (C Ibáñez et al., 2014). In the food industry, metabolomics method has been used for food safety inspection, detection of raw materials and quality of products, the identification of major chemical compounds among species (Sidwick et al., 2017). Multiple metabolomics platforms and technologies have allowed us to substantially enhance the level of metabolome coverage. Using UPLC-QTOF (liquid chromatography quadrupole time-of-flight mass spectrometry) five new compounds (pyrimethanil, 1-palmitoyllysophosphatidylcholine, harman, and norharman, 4-hydroxy-3-methoxycinnamaldehyde) have been described as the characteristic markers that could be used to discriminate coffee obtained by cold-brew, pour-over, or boiled methods (Lao et al., 2019). Gas chromatography mass spectrometry (GC-MS) has been used to identify potential defective markers in coffee seeds (Toci & Farah, 2014). An untargeted metabolomic method based on the mid-infrared Fourier transformed spectroscopy (FT-MIR) technique and combining with chemometrics have been used to identify and quantify adulterants in *Arabica* coffee (Flores-Valdez et al., 2020). Other studies used NMR (nuclear magnetic resonance) to evaluate lipophilic extracts from coffee samples, which can be regarded as a rapid prescreening tool to discriminate between *Arabica* and *Robusta* coffee beans (Monakhova et al., 2015). Moreover, caffeine, trigonelline, chlorogenic acids, sucrose, choline, amino acids and organic acids were used to discriminate species and origins of *Arabica* and *Robusta beans* from six different geographic regions by ^{13}C NMR (F Wei et al., 2011). In addition, High Performance Liquid Chromatography (HPLC)-FLD fingerprinting segments has been recognized as a cheap, simple, feasible, and widely used technique to address coffee characterization, classification, and authentication (Nuñez et al., 2021).

In this study, the quality of 18 coffee bean samples was analyzed by a combination of UHPLC-QE-MS techniques and fingerprint method. UHPLC-QE-MS with detailed assignment information coupled with PCA and OPLS-DA models was applied to identify and characterize metabolites significantly associated with certain cultivars and geographical origins of green coffee beans. This study establishes the foundation for determining special features of coffee beans from different producing regions, as well as for developing the specialty coffee.

2. Materials and methods

2.1. Coffee samples

As shown in Table 1, a total of 18 green coffee bean samples were sourced from importing companies in Chinese and selected to represent multiple origins around the world in 2019. The green coffee beans were frozen at − 20 °C until analyzed. The number of LCZK, DHMS, BSLY, PEML, DBK, DBT green coffee beans are all from Yunnan province, which have the characteristic of strong but not bitter taste, fragrant but not strong, and slightly fruity (Dong et al., 2015). Sumatra coffee has been regarded as the gentleman in coffee with its pleasant slight acidity, which loved by coffee amateurs (Wang et al., 2014).The coffee beans from Costa Rica (GS), Guatemala (WD), Mexico (MX), El Salvador (SEW), Columbia (GL), Rwanda (LWD, LWT), Congo (GG), Ethiopia (AS) and the Independent State of Papua New Guineaare (BBY) are well known as specialty coffee and widely loved by consumers. Brazil (BX) is the world's largest coffee producer, accounting for one-third of global consumption of all grades and varieties of coffee (ICO, 2019).

Table 1
Cultivars and origins of green coffee beans used in the present study.

Number	Cultivars	origin
LCZK	Catimor	Lincang, Zhenkang, Yunnan, China, Asia
DHMS		Dehong, Mangshi, Yunnan, China, Asia
BSLY		Baoshan, Longyang, Yunnan, China, Asia
PEML		Puer, Menglian, Yunnan, China, Asia
DBK		Dali, Binchuan, Yunnan, China, Asia
DBT	Typica	Dali, Binchuan, Yunnan, China, Asia
MT	Arabica	Mandheling, Indonesia, Asia
GS		Costa Rica, North America
WD		Guatemala, North America
MX		Mexico, North America
SEW		El Salvador, North America
GL		Columbia, South America
BX		Brazil, South America
LWD		Rutsiro, Rwanda, Africa
LWT		Rulindo Tumba, Rwanda, Africa
GG		Congo, Africa
AS		Ethiopia, Africa
BBY		The Independent State of Papua New Guinea, Oceania

2.2. Chemicals and reagents

Methanol, Acetonitrile, Ammonium acetate, Acetic acid, Octoic acid and Glutamic acid were purchased from CNW Technologies (Shanghai, China). Water was generated by Merck Millipore (Merck Millipore,USA). Folin-Ciocalteu, DNS were purchased from Sigma, USA. $NaNO_2$, $AlNO_3$, glucose, Na_2CO_3, gallic acid, anthranone-sulfuric were purchased from Tianjin Windship Chemical Century Technology Company.

2.3. Physicochemical analyses

Moisture: The moisture content of the raw beans was determined in triplicate, by drying 1.00 g of green beans in an oven (Marconi, Piracicaba, Brazil) at 105 ± 1 °C until constant weight.

Color: To measure the colour, the green coffee beans were first ground in a mill (Perten) and sieved through an 80-mesh. The CIELAB system was used and values of L* (lightness), a* (red-green component) and b* (yellow-blue component) were obtained directly from the equipment (colour Quest XE-Hunterlab, Reston, USA).

Polyphenol content: Approximately 0.25 g of ground green coffee mixed with 5 mL of 85% methanol, and filtered. The standard and filtrate were taken 0.5 mL and added 0.5 mL of Folin-Ciocalteu reagent was mixed for 5 min. Then 1.0 mL saturated sodium carbonate added and reacted with the samples for 30 min at dark condition. The detection absorbance is 760 nm at spectrometer. A calibration curve was made from gallic acid standard solution (0.00, 0.02, 0.04, 0.06, 0.08 and 0.1 mg/mL) and the blank was prepared with distilled water.

Flavone content: After preparing extracted coffee (the same as Polyphenol content), a calibration curve was made from rutin standard solution (0.00, 0.02, 0.04, 0.06, 0.08 and 0.10 mg/mL). Standard and samples were taken 1.0 mL and added 0.5 mL 5% $NaNO_2$ mixed for 6 min, next added 0.5 mL 10% $AlNO_3$ mixed for 6 min, then added 4 mL 4% NaOH mixed for 10 min at dark condition. The detection absorbance is 510 nm at spectrometer.

Total soluble sugar content: Approximately 1.00 g of ground green coffee mixed with 20 mL of water, and filtered. The standard and filtrate were taken 2 mL and added 6 mL 2 g/L anthranone-sulfuric acid reagent, put it in the boiling water react with 6 min. The detection absorbance is 625 nm at spectrometer after it cool to room temperature. A calibration curve was made from glucose standard solution (0.02, 0.04, 0.06, 0.08 and 0.1 mg/mL) and the blank was prepared for distilled water.

Reducing sugar content: After preparing extracted coffee (the same as total soluble sugar content), the standard and filtrate were taken 2 mL and added 2 mL DNS reagent, put it in the boiling water react for 15 min. The detection absorbance is 540 nm at spectrometer after it cool to room temperature. A calibration curve was made from glucose standard solution (0.00, 0.02, 0.04, 0.06, 0.08 and 0.1 mg/mL).

2.4. Untargeted metabolomics analysis

2.4.1. Metabolites extraction

20 mg of ground green coffee sample was weighted to an EP

(eppendorf) tube after grounded with liquid nitrogen, and 1000 μL extract solution (acetonitrile: methanol: water = 2:2:1) with isotopically-labelled internal standard mixture (octoic acid and glutamic acid) was added. After 30 s vortex, the samples were homogenized at 35 Hz for 4 min and sonicated for 5 min in ice-water bath. The homogenization and sonication cycle was repeated for 3 times. Then the samples were incubated for 1 h at −40 °C and centrifuged at 12000 rpm for 15 min at 4 °C. The resulting supernatant was transferred to a fresh glass vial for analysis. The quality control (QC) sample was prepared by mixing an equal aliquot of the supernatants from all of the samples.

2.4.2. UHPLC-QE-MS analysis

UHPLC-QE-MS analyses were performed using an UHPLC system (Vanquish, Thermo Fisher Scientific) with a UPLC HSS T3 (2.1 mm × 100 mm, 1.8 μm) coupled to Q Exactive HFX mass spectrometer (Orbitrap MS, Thermo). The mobile phase consisted of 5 mmol/L ammonium acetate and 5 mmol/L Acetic acid in water (A) and Methanol (B). The analysis was carried with elution gradient as follows: 0–1.2 min, 1% B; 1.2–9.5 min, 1%–99% B; 9.5–11.8 min, 99% B; 11.8–12.0 min, 99%–1% B; 12–15 min, 1% B. The column temperature was 30 °C. The autosampler temperature was 4 °C, and the injection volume was 2 μL.

The QE HFX mass spectrometer was used for its ability to acquire MS/MS spectra on information-dependent acquisition (IDA) mode in the control of the acquisition software (Xcalibur, Thermo). In this mode, the acquisition software continuously evaluates the full scan MS spectrum. The ESI source conditions were set as following: sheath gas flow rate as 50 Arb, Aux gas flow rate as 10 Arb, capillary temperature 320 °C, full MS resolution as 60000, MS/MS resolution as 7500, collision energy as 10/30/60 in NCE mode, spray Voltage as 4.0 kV (positive) or −3.8 kV (negative), respectively.

2.5. Data processing and statistical analysis

2.5.1. Statistical analysis

Results were analyzed using one-way ANOVA and the IBM SPSS Statistics 22.0, drawn in GraphPad Prism v.8.0.2 software. Data were presented as mean value ± standard deviation. Furthermore, the Duncan honest significant difference test was performed at a significance level of 0.05.

2.5.2. UHPLC-QE-MS data analysis

This analysis was performed at the Biotree Company (Shanghai, China), the raw data were converted to the mzXML format using ProteoWizard and processed with an in-house program, which was developed using R and based on XCMS, for peak detection, extraction, alignment, and integration. Then an in-house MS2 database (BiotreeDB) was applied in metabolite annotation. The cutoff for annotation was set at 0.3.

The UHPLC-QE-MS fingerprint was referred to the method of professional software similarity evaluation system for chromatographic fingerprint of traditional Chinese medicine (Version 2004 A) composed by the Chinese pharmacopoeia committee. The original values of each group of positive and negative ions in the compound identification list were sorted respectively. The compound with the largest peak volume was designated as the reference peak (divided into positive and negative ions), and the other compounds were divided by the reference compound to obtain the ratio with the reference peak, while the common peak was established by the median method (the composition of the common peak is the median value of each compound in the peak volume of each origin). Then, the RSD was calculated for each compound in each region according to the ratio of the reference peaks, calculated the similarity at intervals of n/10 according to the number of compounds (n), using the "angle cosine method" by the R 3.5.1 software.

Multivariate statistical analysis was carried out using SIMCA 14.0 software (MSK Data Analytics Solutions,Umeå, Sweden) where data were centered and divided by the square root of the standard deviation as scaling factor (Pareto scaling). An unsupervised principal component analysis (PCA) was first applied to investigate clustering existing in the analyzed samples. Then, orthogonal partial least squares discriminant analysis (OPLS-DA) was used to discriminate samples according to their areas. The quality of the models was evaluated by the goodness-of-fit parameters R^2X, R^2Y and Q^2. If the standard compounds could be commercially acquired, they were analyzed under identical analytical conditions to obtain their retention times and MS/MS fragmentation patterns in order to confirm the metabolite identity.

3. Results and discussion

3.1. Determination of physiochemical properties of coffee using a targeted approach

The International Coffee Organization establishes a strict limit for water content in coffee beans, i.e. between 8 and 12.5% after processing. A certain moisture content likely contributes to extend food quality or shelf-life. As shown in Fig. 1 (A), moisture content in all coffee beans ranged from 5.72 to 10.93%.

Color is also an important attribute of coffee beans. Previous studies have shown that coffee beans became more pale and yellow when moisture content is low, within an increased in L*, a* and b* values and tolerance to prolonged preserving time (Rendn et al., 2014). Colour changes in raw beans are one of the main parameters that affect the quality of coffees, since heterogeneous colour distribution indicates problems during processing and storage conditions (Wintgens, 2004). In the present study, color of evaluated coffee samples was yellow-green, as well as bright ($L^* > 0$), $a^* > 0$ tending to wards negative axis, and $b^* > 0$ (Fig. 1 B,C, D). When coffee samples were divided into five groups according to their origin, significant differences in L* values were found coffee beans from Asia, Africa, and Oceania ($P < 0.05$).

Green coffee beans are rich in phenolic acids, which are an ingredient holding relevant biological activity and responsible for conferring the characteristic coffee flavor (Wang et al., 2017). 4,5-Di-O-caffeoylquinic acid, chlorogenic acid and cryptochlorogenic acid are the most important phenolic acids in coffee, which have been known to have antioxidant and anti-inflammatory activity, and other health-promoting properties for humans (Gilberto et al., 2019; Poisson et al., 2017). In addition, since carbohydrates are the main precursors of coffee flavor, they are also important in the evaluation of the quality of coffee beans (Zhang et al., 2015). Heterocyclic compounds are the most important components providing the distinctive flavor of roasted coffee bean, which are generally formed from precursors such as polysaccharides, lipids, proteins, and free amino acids (Lee & Shibamoto, 2010). In Fig. 1 (E, F, G, H), it is reported the results of the physiochemical characterization of coffee samples in terms of their content of polyphenols, flavones, total soluble sugar, and reducing sugar. The content of polyphenols and flavones were significantly higher in coffee beans from Baoshan Longyang of latitude 25.08° North and longitude 99.10° east compared to other locations ($P < 0.05$). The content of soluble sugar in Dehong Mangshi of latitude 24.50° North and longitude 98.50° east and the content of reducing sugar in El Salvador of latitude 9.03° North and longitude 38.42° east were significantly higher than coffee from other locations ($P < 0.05$). Moreover, considering separation of coffee samples into 5 groups by their geographical origin, the flavonoid content of Asian coffee beans was significantly higher compared to coffee from other continents ($P < 0.05$). The content of reducing sugar in coffee beans from South and North America was significantly higher than coffee harvested in Oceania ($P < 0.05$).

Phenolic acids, sugars, flavonoid, proteins, free amino acids, fatty acids are the principal flavor precursor compounds in green coffee, which formed desirable aroma of coffee during the roasting step at high temperature beyond 200°C (Flament, 2002; Illy and Viani, 1995). Poisson L et al. (Poisson et al., 2009) found that the entire penetration of aqueous solutions of sucrose into the green coffee bean within 5–6 h.

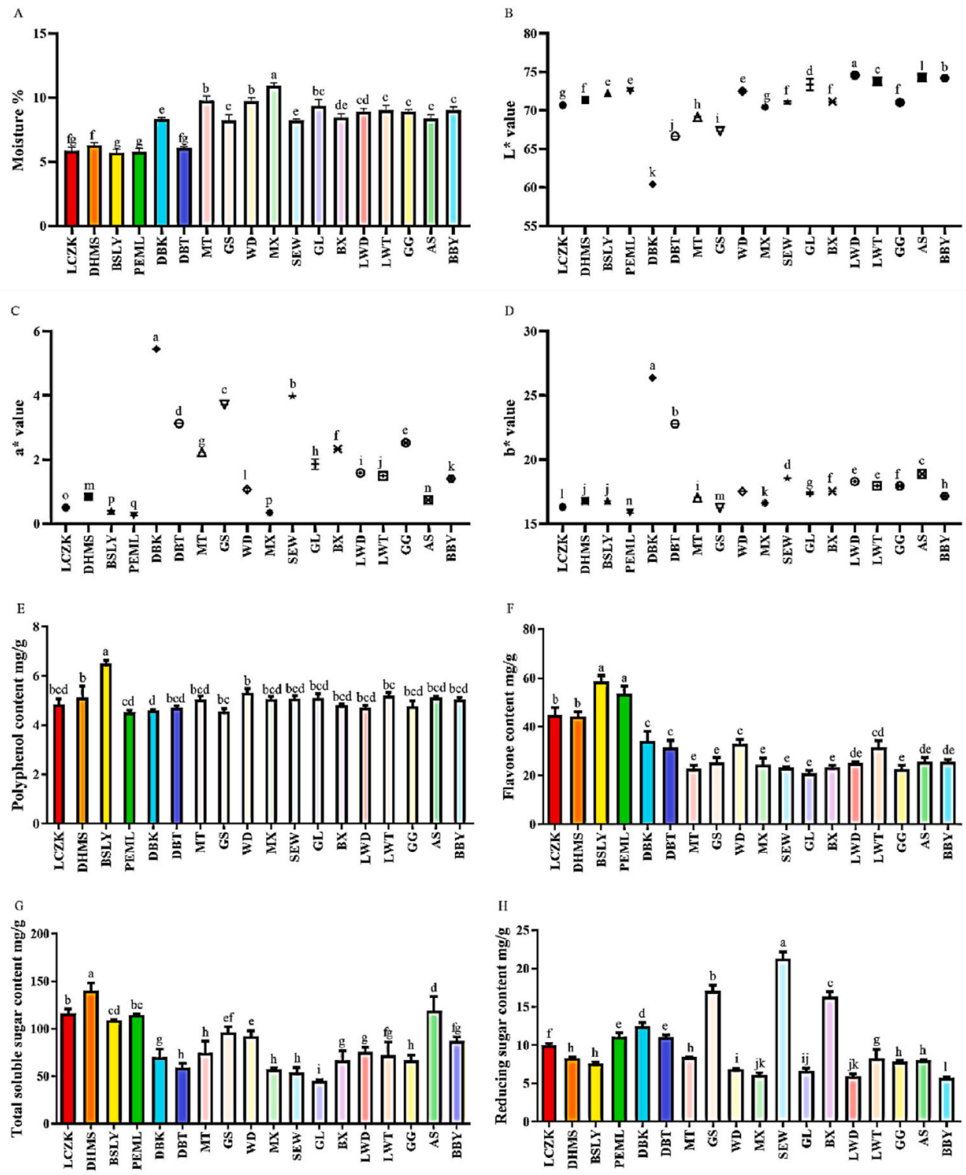

Fig. 1. Variation in physiochemical properties of moisture content (A), color difference (B, C, D), polyphenol content (A), flavone content (B), total soluble sugar content (C), reducing sugar content (D) of different region coffee beans. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Elhalis H et al. (Elhalis et al., 2020) discovered that wet fermentation of coffee beans can significantly enhance quality of the coffee beverage, flavor, aroma and sensory by directly increasing the levels of key precursors such as organic acids, esters, alcohols, and aldehydes. Different processing methods to obtain green coffee beans from coffee cherries, namely, the wet, dry, and honey process, can producing diferent enzymes (pectinase), acids, and alcohols. This lead to different impact on the sensory quality of the coffee (Haile and Kang, 2019). So this provides a theoretical basis for studies to improve coffee flavour.

3.2. Metabolomics of coffee using an untargeted UHPLC-QE-MS based approach

3.2.1. Metabolomics analysis of three cultivars of coffee beans

Collectively, 18622 features were extracted from the total ion chromatogram from UPLC-QE-MS analysis in positive ion mode, and 12618 features were obtained in negative mode. Principal component analysis (PCA) and orthogonal partial least squares discriminant analysis (OPLS-DA) were performed to differentiate coffee samples. PCA scores plots of coffee samples reveal separation of coffees into distinctive clusters of groups (Fig. 2 (A, B)). Principal component 1 (PC1) and PC2 contributed more to variance, explaining 23.8% and 8.9%, of variability among compounds extracted from positive ion mode, and 26.5% and 10.5% among compounds from negative ion mode, respectively. Furthermore, the presence of overlapping samples in them indicates more similarities in composition and concentration of molecules in the coffee samples, whereas similar molecular composition and concentration may result in aggregation of samples corresponding to the same coffee species or from different species but with similar metabolic composition. Therefore, within-group error caused by the unsupervised PCA method must be taken into consideration. In order to eliminate the random error, a supervised method was used to discriminate group samples.

OPLS-DA analysis was used to distinguished coffee samples from different cultivars. Coffee samples were divided into three groups (*Catimor, Typica* and *Arabica*) according to their cultivars, which revealed that the three groups were grouped separately regardless of the ionization mode. Potential markers were chosen based on their

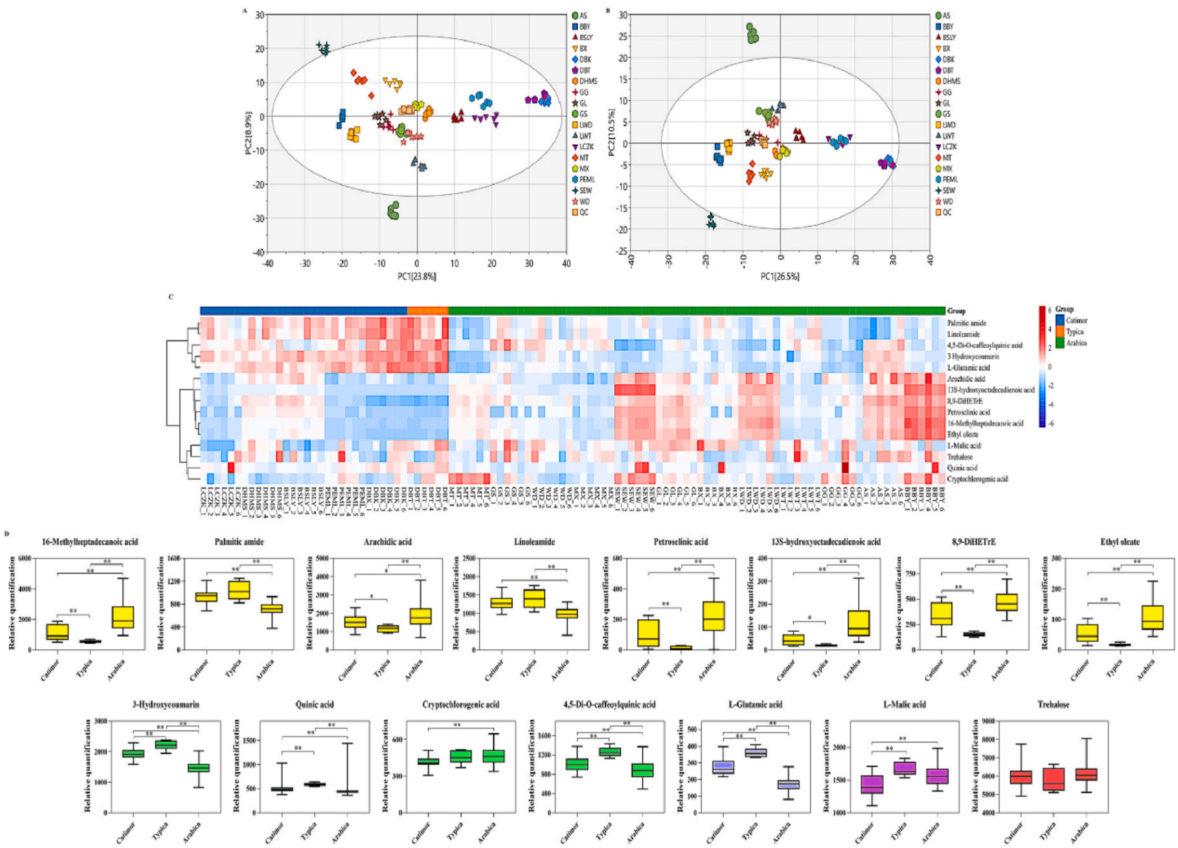

Fig. 2. The PCA scores plots of compound abundances obtained in positive ion mode (A) and negative ion mode (B) from coffee samples. (C) Heatmaps of three varieties of coffee beans. (D) The relative count of 15 different compounds (include 16-methylheptadecanoic acid, palmitic amide, arachidic acid, linoleamide, petroselinic acid, 13S-hydroxyoctadecadienoic acid, 8,9-DiHETrE, ethyl oleate, 3-hydroxycoumarin, quinic acid, cryptochlorogenic acid, 4,5-di-o-caffeoylquinic acid, L-glutamic acid, L-malic acid, trehalose).

relevance to establishing variation and correlation within the dataset. In addition, heatmaps were used to highlight significant differences between groups.

The heatmaps display the data with the cultivars on the horizontal axis and the compounds on the vertical axis, and the correlation between varieties is determined independently by the abundance of compound. In total, 15 different compounds were filtered using variance results and which were selected as potential markers to discriminate coffee cultivars (Fig. 2 (C)). The relative contents of 3-hydroxycoumarin, quinic acid, 4,5-di-o-caffeoylquinic acid, cryptochlorogenic acid (4-O-caffeoylquinic acid), palmitic amide, linoleamide, arachidic acid, 16-methylheptadecanoic acid, ethyl oleate, 13S-hydroxyoctadecadienoic acid, petroselinic acid, 8,9-DiHETrE, L-malic acid, trehalose, L-glutamic acid varied greatly among the three coffee varieties evaluated in this study. It is well established that *Catimor*, *Typica* and *Arabica* are closely related coffee plants with sharing a high degree of genetic similarity. *Typica* serves as the genetic basis for all *Arabica* variants or genetic screens, whereas *Catimor* is a hybrid of the *Caturra* (*Bourbon* branch of the *Arabica*) and *Robusta* (Filho and Carvalho, 1957; Zhang et al., 2018).

Morover, the relative content of 3-hydroxycoumarin, quinic acid, 4,5-di-o-caffeoylquinic acid, palmitic amide, linoleamide, L-malic acid, L-glutamic acid were significantly higher in *Typica* coffee beans compared with other evaluated varieties (see Fig. 2 (D)). At the same time, the trend of *Arabica* variation were contrary to that of *Typica*, likely as a result of the conditions related to altitude (3248 m above sea level) and mild temperatures (suitable temperature is 24–29°C) of *Typica* cultivation. And the coffee samples were located in the southwest region of China, in low latitudes plateau monsoon climate zone that has a moderate temperature and abundant rainfall (Zhao et al., 2015) In contrast, *Arabica* coffee is usually grown at plateau areas (600–2000 m above sea level) in cooler climates (suitable temperature is 15–24 °C). Other studies employed chemometric tools to separate *Arabica* coffee cultivars into genealogical groups based on the chemical composition of raw coffee beans, which revealed that arachidic acid and stearic acids can be considered markers for the *Bourbon* cultivar; palmitoleic and oleic acids, total sugars and the protein content for the *Timor Hybrid* cultivar; and myristic and linoleic acids and sucrose for the exotic genotypes group (Malta et al., 2020).

3.2.2. *Metabonomics analysis of coffee from five continents*

Supervised orthogonal partial least squares discriminant analysis (OPLS-DA) was used to filter noise, improve analytical capability and effectiveness of the model, eliminate sampling errors and other random errors, and maximize differences among evaluated groups. Firstly, coffee beans were divided into five groups according to their origin: Asia, north America (NA), south America (SA), Africa, and Oceania.

In Fig. 3 reach data point in the S-plot represents a metabolic compound. S-plot can be a useful tool to identify biochemical variables, as reflected by the values of covariance and correlation between each variable and the component of the model, which denotes, respectively, the contribution of variables in the model and the weighting coefficient of variables. Within each group, the farther the metabolite deviates from zero, the greater its contribution for classification of coffee, and more likely its relevance to be use potential marker. According to the importance of variables as show in the S-plot, 139 different compounds were filtered using absolute VIP values greater than 1.0. The differential metabolites between different groups were categorized according to their class, which included polyphenols, alkaloids, fatty acyls, amino acids, alcohols and polyols, polysaccharides, as well as the organic acid. And the majority that were classifiable were found to be fatty acyls,

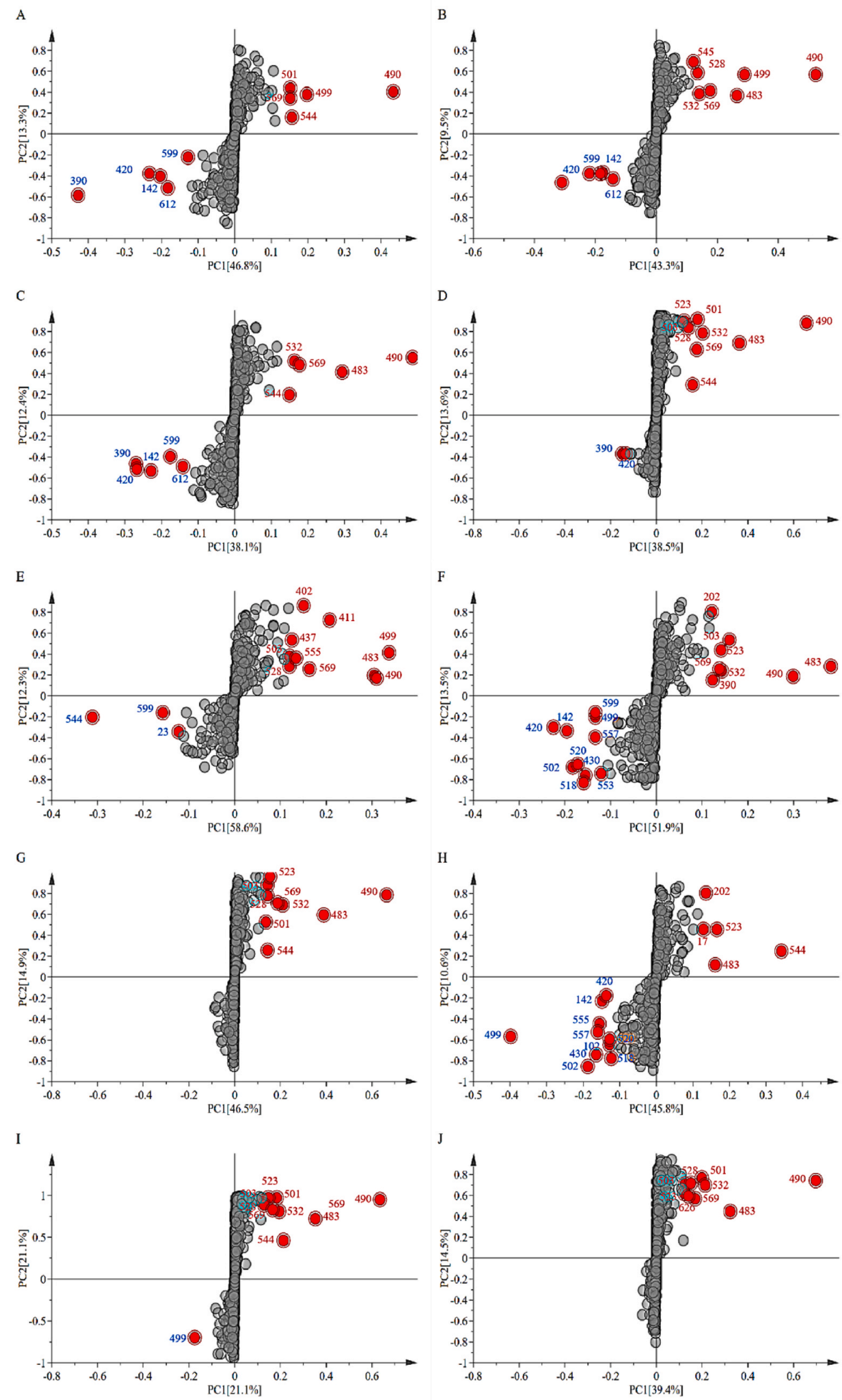

Fig. 3. S-plots of five continents groups. (A: Asia vs North America, B: Asia vs South America, C: Asia vs Africa, D: Asia vs Oceania, E: North America vs South America, F: North America vs Africa, G: North America vs Oceania, H: South America vs Africa, I: South America vs Oceania, J: Africa vs Oceania).

amino acids and their related metabolites (Fig. 4 (A)). The relative count of arachidic acid (483), 16-methylheptadecanoic acid (490), 13S-hydroxyoctadecadienoic acid (501), 12-methyltridecanoic acid (503), ethyl oleate (528) and petroselinic acid (532) varied greatly among the coffee samples from all five continent evaluated in this study (Fig. 4 (B)).

In order to discriminate the change of differential metabolites among the continents, *t*-test ($P < 0.05$) were be used, we identified 16 differential metabolites (Fig. 4 (C)). It is evident that the relative quantification of chlorogenic acid, 3-hydroxycoumarin, 4,5-Di-O-caffeoylquinic acid, linoleamide, palmitic amide, L-glutamic acid, D-aspartic acid and L-phenylalanine were significantly more abundant in Asia coffee samples than in other continent. Cryptochlorogenic acid (4-O-caffeoylquinic acid), arachidic acid were significantly higher in Oceania coffee samples than in other continent. Alanine was higher in north America coffee samples. And the relative quantification of organic acid such as succinic acid, L-malic acid were both higher in north America and south America. These components are known to have antioxidant activity and various beneficial properties in human health (Lecoultre et al., 2014; Yoshinari & Igarashi, 2015). The changes of green coffee bean composition is likely due to geographic origin, environmental factors in pre-harvest processing, presence of defective beans, and ripening stage (Toledo et al., 2016). This observation is in alignment with the above-discussed findings that the polyphenols content varied to great extent in coffee samples from different continents. Previous study reported that the higher levels of sucrose and the lower levels of quinic acid, choline, acetic acid, γ-aminobutyric acid (GABA), and fatty acids observed in high-grade green coffee beans, compared to commercial-grade beans, were suggestive to be multiple markers for characterizing specialty green coffee bean (Kwon et al., 2015).

3.2.3. Comparative metabonomic analysis of coffee from Yunnan and other countries

The province of Yunnan in China is a well-known coffee producing region that has three climatic subtypes (subtropical and tropical monsoon climate, and plateau mountain climate), areal differentiation and vertical changes, which allow for a rich diversity of flavor and quality of coffee harvested locally. Therefore, Yunnan coffee was chosen as a criterion for comparison with coffee harvested in other countries to determine the characteristic components of *Catimor, Typica,* and *Coffea arabica* L.. As shown in Fig. 5 (A, B), 94 different distinct features were found, among which, 17 compounds could be considered as markers that allow differentiation of coffee beans from Yunnan from that produced in other regions; these included quinic acid, 4,5-di-O-caffeoylquinic acid, cryptochlorogenic acid, 3-hydroxycoumarin, oleamide, palmitic amide, linoleamide, arachidic acid, 13S-hydroxyoctadecadienoic acid, 8,9-DiHETrE, petroselinic acid, 16-methylheptadecanoic acid, ethyl oleate, melibiose, trehalose, L-glutamic acid, L-malic acid. It is evident that the relative quantification of 3-hydroxycoumarin, 4,5-Di-O-caffeoylquinic acid, quinic acid, linoleamide, palmitic amide, melibiose, L-glutamic acid were significantly more abundant in China coffee samples than in other countries. Hu, Peng, Wang, et al. (2020) found that 5-CQA (5-dicaffeoylquinic acids), sugars, caffeine and trigonelline could be considered as markers to distinguish the maturation stage of *Coffea arabica* L. In addition, previous research revealed that quinic acids, malate, sugars, trigonelline, γ-butyro-lactone, and acetate could potentially be used as indicators of fresh roasting *Arabica* (*Catimor*) cultivar (Hu, Peng, Gao, et al., 2020).

3.2.4. UHPLC-QE-MS fingerprint analysis of coffee beans

UPLC-QE-MS fingerprint analysis of coffee beans was performed to enable visualization of homogeneity and stability of complex components in quality (Peishan Xie, 2001). UPLC-QE-MS chromatograms of coffee beans samples (18 districts) were shown in Supplementary Fig. 1 (A, B). Correlation coefficient was used to estimate similarity of chromatographic fingerprints of coffee beans and the reference chromatogram, and a representative standard chromatographic fingerprint for a group of chromatograms was determined. Correlation coefficients of coffee beans for 78 compounds were greater than 0.952, which indicate different internal qualities. These identified compounds were classified into seventeen different polyphenols and seventeen different fatty acyls, alkaloids (7 compounds), terpenoids (6 compounds), sugars (5 compounds), amino acids (4 compounds), organic acids (1 compounds), and other compounds (21 compounds) (Supplementary Table 1). Differences among these variables could be considered as unique fingerprints of evaluated coffee samples, as well as can provide more comprehensive information for quality control and identification of potential markers of coffee beans from different regions. Among identified potential markers are included polyphenols (3-hydroxycoumarin, 4,5-di-O-caffeoylquinic acid, and quinic acid), which have promising effects in disease prevention (Araújo et al., 2020). Palmitic amide, linoleamide, oleamide are fatty acyls which bring subtle and pleasant flavors to coffee and also constitute the main compounds affecting texture and taste of coffee-based drinks (Oestreich-Janzen, 2010). Melibiose, trehalose belong to carbohydrate, affects flavor of coffee and also has anti-inflammatory properties (Yohei et al., 2016). Wei F et al. (Wei et al., 2011) employed ^{1}H–^{13}C NMR to analyze roasted *Arabica* coffee beans from Colombia and found that 3-CQA, 4-CQA, 5-CQA, quinic acid, choline, myo-inositol, and malic acid could be regarded as markers for the analysis of complex mixture without prior separation.

3.3. Determination of key metabolites in coffee

Collectively, results obtained in UHPLC-QE-MS fingerprint analysis, PCA analysis, and OPLS-DA models showed that 10 compounds (3-hydroxycoumarin, 4,5-di-O-caffeoylquinic, cryptochlorogenic acid, palmitic amide, linoleamide, arachidic acid, petroselinic acid, trehalose, L-glutamic acid, L-malic acid) could be regarded as characteristic markers for coffee harvested in 18 regions worldwide, which are in line with previously published data.

Other studies used an untargeted metabolomics multiplatform which is based on the integration of three orthogonal techniques, suggesting that 4,5-di-O-caffeoylquinic, quinic acid, chlorogenic acid, and choline could tentatively be potential biomarkers (Marina Alegre et al., 2020). Interestingly, 4,5-di-O-caffeoylquinic, quinic acid, and chlorogenic acid have been previously described as characteristic markers of *Arabica* coffee (Pérez-Míguez et al., 2018). Among the studied, the relative content of phenolics is quite rich in Yunnan coffee, with consistency in this study.

Organic acids (such as citric acid and malic acid) are the essential factor affecting the sensory properties, and abundance of this group contribute to the acidity of coffee (Ribeiro et al., 2018). Organic acids also enhance the characteristic fruity flavor in brewed coffee, and the differences in content and proportion distribute the uniqueness of different coffees (Wang et al., 2021). Citric acid and malic acid readily bound to sugars, which reduce the taste of tart and produce the similar taste of wine and also enhance the fleshiness of roasted coffee. The relative count of malic acid was higher in coffee from south American (*Arabica*) and *Typica.*

4. Conclusions

Previous metabolomics studies of coffee have focused mostly on roasted coffee beans or on coffee from a particular geographical region. Furthermore, only few studies have been performed on green coffee beans from different countries, producing regions and species. Thus, an untargeted metabolomic approach based on UHPLC-QE-MS combined with fingerprint analysis revealed an effective approach to enable differentiation of coffee beans from 18 regions different geographical origins. PCA analysis of raw LC/MS data in both positive and negative ion modes enabled separation of coffee samples into score plots. Furthermore, an OPLS-DA model was built to filter the markers, and ten potential markers were putatively annotated. To the best of our

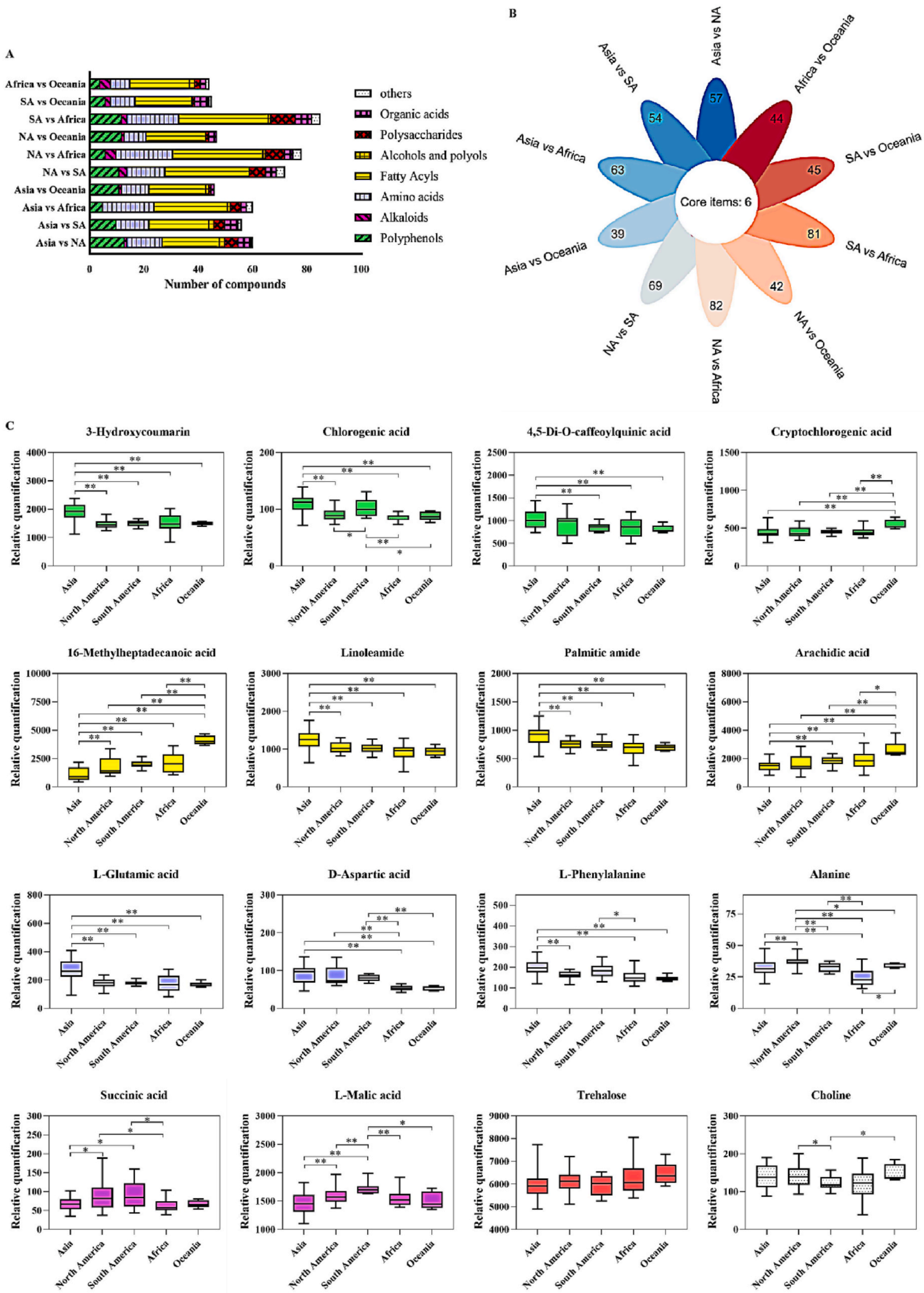

Fig. 4. The number of compounds of five continents groups (A). Venn diagram displaying the total number of different metabolites tentatively identified by the untargeted multiplatform (N A: North America, S A: South America)(B). (C)The relative count of 15 different compounds (include 3-hydroxycoumarin, chorogenic acid, cryptochlorogenic acid, 4,5-di-o-caffeoylquinic acid, 16-methylheptadecanoic acid, palmitic amide, arachidic acid, linoleamide, petroselinic acid, L-glutamic acid, D-aspartic acid, L-phenylalanine, alanine, succinic acid, L-malic acid, trehalose, choline).

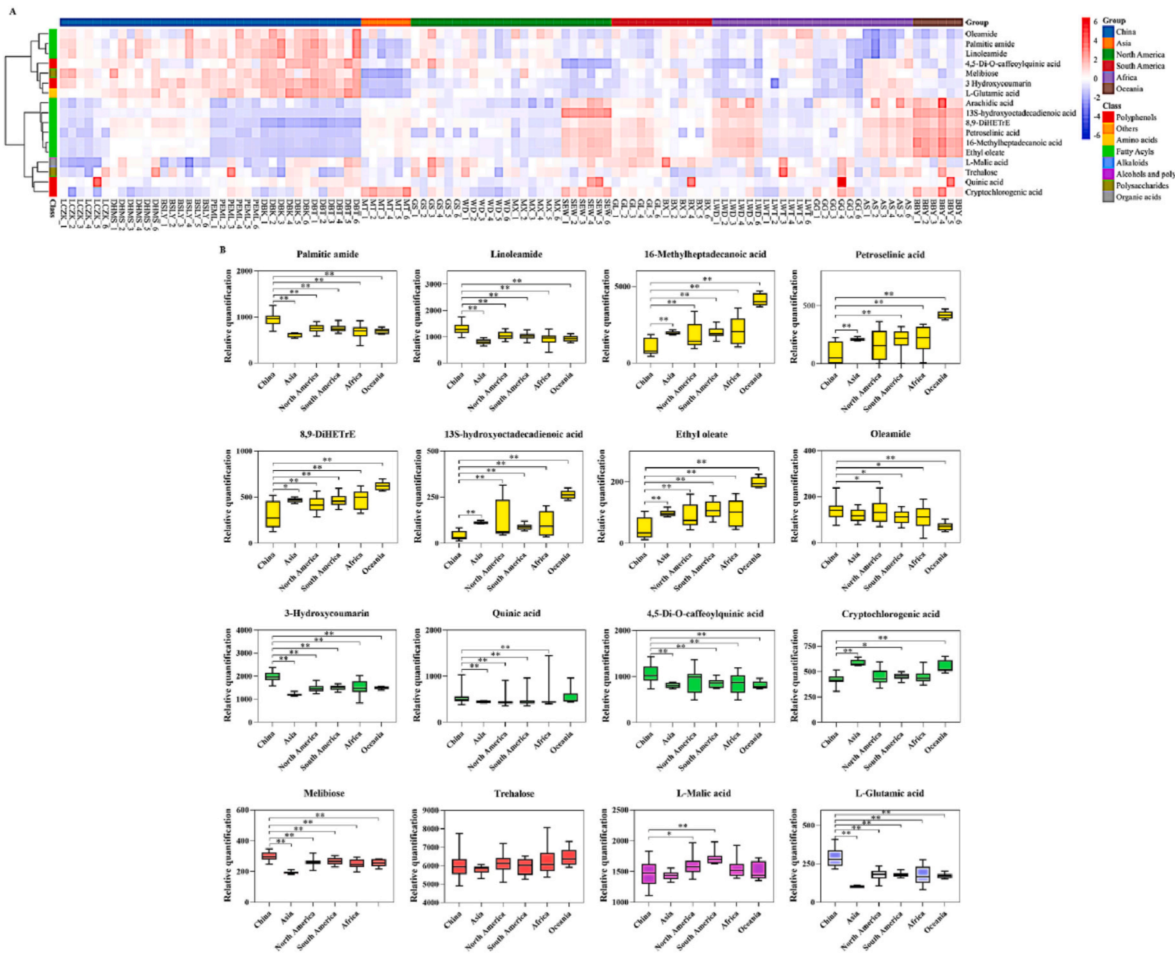

Fig. 5. Heatmaps from hierarchical clustering analysis of compound abundances obtained in positive ion mode and negative ion mode from 18 regions of coffee samples (A). (B) The relative count of 15 different compounds (include 16-methylheptadecanoic acid, palmitic amide, arachidic acid, linoleamide, petroselinic acid, 13S-hydroxyoctadecadienoic acid, 8,9-DiHETrE, ethyl oleate, oleamide, 3-hydroxycoumarin, quinic acid, 4,5-di-o-caffeoylquinic acid, cryptochlorogenic acid, melibiose, trehalose, L-malic acid, L-glutamic acid).

knowledge, this is the first report comprising the use of fingerprint methods for the characterization of coffee and the first description of 3-hydroxycoumarin in coffee beans. Ten different families of compounds were considered as potential markers of the coffee beans: 3-hydroxycoumarin, 4,5-di-O-caffeoylquinic, cryptochlorogenic acid, palmitic amide, linoleamide, arachidic acid, petroselinic acid, trehalose, L-glutamic acid, L-malic acid. Collectively, the developed fingerprint methodology presented herein can be considered as a useful and effective strategy for obtaining information on the metabolome of coffee samples, which is complementary to other methods commonly used in metabolomics. Furthermore, the findings presented in this study reveal that UHPLC-QE-MS is a suitable tool for the design of discrimination strategies in food origin tracing.

Author statement

Conceptualization, M.Y.; Methodology, M.Y.; Software, M.Y.; Validation, M.Y.; Formal analysis, M.Y.; Investigation, Z.F.; Resources, W.Q.; Data curation, Z.C. and L.Z; Writing—Original draft preparation, M.Y.; Writing—Review and editing, M.Y.; Visualization, T.C., L.Z. and P. C.; Supervision, G.J.; Project administration, G.J.; Funding acquisition, G.J. All authors have read and agreed to the published version of the manuscript.

Declaration of competing interest

The authors declare no competing financial interests.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.fbio.2022.101561.

References

Araújo, F. F. D., Farias, D., Neri-Numa, I. A., & Pastore, G. M. (2020). Polyphenols and their applications: An approach in food chemistry and innovation potential. *Food Chemistry, 338*.

Cong, S., Dong, W., Zhao, J., Hu, R., Long, Y., & Chi, X. (2020). Characterization of the lipid oxidation process of Robusta green coffee beans and shelf life prediction during accelerated storage. *Molecules, 25*(5), 1157. https://doi.org/10.3390/molecules25051157

Dam, R., Hu, F. B., & Willett, W. C. J. (2020). *Coffee, Caffeine, and Health, 383*(4), 369–378.

Dong, W., Zhang, F., Zhao, J., Gu, F., & Lu, M. (2015). Application of electronic nose system coupled with HS-SPME-GC/MS for characterization of aroma fingerprint of

roasted coffee beans from different cultivation regions in Yunnan province. *Chinese Journal of Topical Crops, 36*(10), 9.
Elhalis, H., Cox, J., Frank, D., & Zhao, J. (2020). The role of wet fermentation in enhancing coffee flavor, aroma and sensory quality. *European Food Research and Technology*, (2), 1–14. FAO Crop production by country. [on 2020 November 31th]. Available from http://www.fao.org/faostat/en/#data.
Filho, H. A., & Carvalho, A. (1957). Genetics of coffea, 16: The variety typica as a standard for genetical studies of coffea arabica. *Journal of Heredity, 48*(3), 110–117.
Flament, I. (2002). *Coffee flavour chemistry*. Chichester, U.K.: Wiley.
Flores-Valdez, M., Meza-Márquez, O. G., Osorio-Revilla, G., & Gallardo-Velázquez, T. (2020). Identification and quantification of adulterants in coffee (coffea arabica L.) using FT-MIR spectroscopy coupled with chemometrics. *Foods, 9*(7), 851.
Haile, M., & Kang, W. H. (2019). The role of microbes in coffee fermentation and their impact on coffee quality. *Journal of Food Quality*, (12).
Hu, G., Peng, X., Gao, Y., Huang, Y., & Qiu, M. (2020). Effect of roasting degree of coffee beans on sensory evaluation: Research from the perspective of major chemical ingredients. *Food Chemistry, 331*, 127329.
Hu, G., Peng, X., Wang, X., Li, X., & Qiu, M. (2020). Excavation of coffee maturity markers and further research on their changes in coffee cherries of different maturity. *Food Research International, 132*(2), 109–121. ICO Global Coffee Production to Decrease in 2019/20. [on 2019 November 5th]. Available from https://www.ico.org/show_news.asp?id=706.
Ibáñez, C., García-Cañas, V., Valdés, A., & Simó, C. (2014). *Chapter 10 - direct mass spectrometry-based approaches in metabolomics*. Elsevier Science & Technolog. https://doi.org/10.1016/B978-0-444-62651-6.00010-6
Illy, A., & Viani, R. (1995). *Espresso coffee: The chemistry of quality*. London, U.K.: Academic Press.
Jaquet, M., Rochat, I., Moulin, J., Cavin, C., & Bibiloni, R. (2009). Impact of coffee consumption on the gut microbiota: A human volunteer study. *International Journal of Food Microbiology, 130*(2), 117–121.
Kwon, D. J., Jeong, H. J., Moon, H., Kim, H. N., Cho, J. H., Lee, J. E., Hong, K. S., & Hong, Y. S. (2015). Assessment of green coffee bean metabolites dependent on coffee quality using a 1H NMR-based metabolomics approach. *Food Research International, 67*(jan), 175–182.
Lao, F., Xu, Z., Wang, X., Chen, F., Liao, X., Chen, A., & Yang, S. (2019). Use of liquid chromatography quadrupole time-of-flight mass spectrometry and metabolomic approach to discriminate coffee brewed by different methods. *Food Chemistry, 286* (JUL.15), 106–112.
Lara-Guzmán, O. J., Medina, S., Álvarez, R., Oger, C., Durand, T., Galano, J.-M., & Muñoz-Durango, K. (2020). Oxylipin regulation by phenolic compounds from coffee beverage: Positive outcomes from a randomized controlled trial in healthy adults and macrophage derived foam cells. *Free Radical Biology and Medicine, 160*, 604–617.
Larissa, D. O. F., Marcelo, M., Gladyston, C., Gilberto, L., Priscilla, D. L., & Carlos, P. (2016). Sensory description of cultivars (coffea arabica L.) resistant to rust and its correlation with caffeine, trigonelline, and chlorogenic acid compounds. *Beverages, 2* (1), 12.
Lecoultre, V., Carrel, G., Egli, L., Binnert, C., Boss, A., Macmillan, E. L., Kreis, R., Boesch, C., Darimont, C., & Tappy, L. (2014). Coffee consumption attenuates short-term fructose-induced liver insulin resistance in healthy men. *American Journal of Clinical Nutrition, 99*(2), 268–275.
Lee, K. G., & Shibamoto, T. (2010). Analysis of volatile components isolated from Hawaiian green coffee beans (Coffea arabica L.). *Flavour and Fragrance Journal, 17* (5), 349–351.
Malta, M. R., Fassio, L. O., Liska, G. R., Carvalho, G. R., & Pereira, R. G. F. A. (2020). *Discrimination of genotypes coffee by chemical composition of the beans: Potential markers in natural coffees* (p. 109219). Food Research International.
Monakhova, Y. B., Ruge, W., Kuballa, T., Ilse, M., Winkelmann, O., Diehl, B., & Lachenmeier, D. W. (2015). Rapid approach to identify the presence of Arabica and Robusta species in coffee using 1H NMR spectroscopy. *Food Chemistry, 182*(sep.1), 178–184.
Núez, N., Martínez, C., Saurina, J., & Núez, O. (2021). High-performance liquid chromatography with fluorescence detection fingerprints as chemical descriptors to authenticate the origin, variety and roasting degree of coffee by multivariate chemometric methods. *Journal of the Science of Food and Agriculture, 101*(1).
Oestreich-Janzen, S. (2010). Chemistry of coffee. *Comprehensive Natural Products II, 8*(9), 1085–1117.
Pérez-Míguez, R., Castro-Puyana, M., Sánchez-López, E., Plaza, M., & Luisa Marina, M. (2020). Untargeted HILIC-MS-based metabolomics approach to evaluate coffee roasting process: Contributing to an integrated metabolomics multiplatform. *Molecules, 25*(4).
Pérez-Míguez, R., Sánchez-López, E., Plaza, M., Castro-Puyana, M., & Marina, M. L. (2018). A non-targeted metabolomic approach based on reversed-phase liquid chromatography–mass spectrometry to evaluate coffee roasting process. *Analytical and Bioanalytical Chemistry*.
Poisson, L., Auzanneau, N., Mestdagh, F., Blank, I., & Davidek, T. (2017). New insight into the role of sucrose in the generation of α-diketones upon coffee roasting. *Journal of Agricultural and Food Chemistry*. acs.jafc.6b04849.
Poisson, L., Schmalzried, F., Davidek, T., Blank, I., & Kerler, J. (2009). Study on the role of precursors in coffee flavor formation using in-bean experiments. *Journal of Agricultural and Food Chemistry, 57*(21), 9923–9931.
Rendn, M. Y., Salva, T., & Bragagnolo, N. (2014). Impact of chemical changes on the sensory characteristics of coffee beans during storage. *Food Chemistry, 147*(mar.15), 279–286.
Ribeiro, D. E., Borém, F. M., Nunes, C. A., Alves, A., & Dias, L. (2018). Profile of organic acids and bioactive compounds in the sensory quality discrimination of arabica coffee. *Coffee Science, 13*(2), 187.
Sidwick, K. L., Johnson, A., Adam, C. D., Pereira, L., & Thompson, D. F. (2017). The use of liquid chromatography quadrupole time-of-flight mass spectrometry and metabonomic profiling to differentiate between normally slaughtered and dead on arrival poultry meat. acs.analchem *Analytical Chemistry*, 7b02749. acs.analchem.
Toci, A. T., & Farah, A. (2014). Volatile fingerprint of Brazilian defective coffee seeds: Corroboration of potential marker compounds and identification of new low quality indicators. *Food Chemistry, 153*, 298–314.
Toledo, P., Pezza, L., Pezza, H. R., & Toci, A. T. (2016). Relationship between the different aspects related to coffee quality and their volatile compounds. *Comprehensive Reviews in Food Science and Food Safety, 15*(4).
Gilberto, V., Pereira, de M, Dão, P., & de, C. (2019). Exploring the impacts of postharvest processing on the aroma formation of coffee beans – a review. *Food Chemistry, 272* (JAN.30), 441–452.
USDA Coffee. [on 2021 January 26]. Available from https://www. nass. usda. gov/ Statistics _ by _ State/Hawaii/Publications/Fruits _ and _ Nuts/Coffee -01 -26-2021. pdf.
Verhage, F., Anten, N., & Sentelhas, P. C. (2017). Carbon dioxide fertilization offsets negative impacts of climate change on Arabica coffee yield in Brazil. *Climatic Change, 144*(4), 671–685.
Vitaglione, P., Morisco, F., Mazzone, G., Amoruso, D., & D`Argenio, G. (2010). Coffee reduces liver damage in a rat model of steatohepatitis: The underlying mechanisms and the role of polyphenols and melanoidins. *Hepatology, 52*(5), 1652–1661.
Wang, Y., Liao, X., Yao, M., Wang, S., & Gao, H. (2014). Study on the character of volatile flavor of naturaI coffee based on sensory evaluation and electronic nose technology. *Science and Technology of Food Industry, 35*(14), 4.
Wang, Y., Wang, X., Hu, G., Hong, D., & Qiu, M. (2021). Chemical ingredients characterization basing on 1H NMR and SHS-GC/MS in twelve cultivars of coffea arabica roasted beans. *Food Research International, 147*(7), 110544.
Wang, L., Yao, J., & Chaomei, M. A. (2017). Advances in research on chlorogenic acid. *Journal of Food Science and Biotechnology*.
Wei, F., Furihata, K., Hu, F., Miyakawa, T., & Tanokura, M. (2011). Two-dimensional 1H-13C nuclear magnetic resonance (NMR)-based comprehensive analysis of roasted coffee bean extract. *Journal of Agricultural and Food Chemistry, 59*(17), 9065.
Wintgens, J. N. (2004). *Coffee: Growing, processing, sustainable production: Coffee: Growing, processing, sustainable production: A guidebook for growers, processors, traders, and researchers*.
Xie, P. (2001). Concept, attribute, technology and application of chromatographic fingerprint identification of traditional Chinese Medicine. *Chinese Journal of traditional Chinese Medicine, 26*(10), 653–655.
Yohei, S., Mitsuru, T., Moe, K., Chiseko, S., Takako, H., Wataru, H., & Toshiiyuki, N. (2016). Enzymatic synthesis of novel oligosaccharides from N-acetylsucrosamine and melibiose using Aspergillus Niger α-galactosidase, and properties of the products. *Bioscience Biotechnology & Biochemistry, 80*(9), 1836–1842. Bioscience, Biotechnology, and Biochemistry: Vol 80, No 9.
Yoshinari, O., & Igarashi, K. (2015). *Coffee in health and disease prevention || antidiabetic effects of trigonelline* (pp. 765–775).
Zhang, feng, Dong, wenjiang, Wang, kaili, Gu, F., Xiong, shanbai, & Zhao, jianpin (2015). HS-SPME/GC-MS analysis of volatile components of coffee beans roasted in different regions of Yunnan. *Food Industry Technology, 36*(11), 273–280.
Zhang, H., Guo, T., Kuang, Y., Li, J., Bai, X., Xiao, B., Xia, H., Zhou, H., Lin, X., & Xiao, Z. (2018). The influence factors on catimor cupping quality and its promotion measures. *Tropical Agricultural Science & Technology, 41*(1), 7.
Zhao, Y., Zhao, J., Zhao, C., Zhou, H., Li, Y., Zhang, J., Hu, C., Li, W., & Peng, X. (2015). A metabolomics study delineating geographical location-associated primary metabolic changes in the leaves of growing tobacco plants by GC-MS and CE-MS. *Scientific Reports, 5*, 16346.